

DIVISION

Dividing one number by another often gives a decimal answer. Sometimes it is easier to turn decimals into whole numbers before dividing them.

Short division with decimals

Many numbers do not divide into each other exactly. If this is the case, a decimal point is added to the number being divided, and zeros are added after the point until the division is solved. Here 6 is divided by 8.

Both numbers are whole. As 8 will not divide into 6, put in a decimal point with a 0 after it and carry the 6.

add a decimal point on the answer line
8 goes into 60 7 times with 4 left over
carry 6
add a 0 after the decimal point
add a decimal point after 6

Dividing 60 by 8 gives 7, with a remainder of 4. Write the 7 on the answer line, add another 0 after the decimal place, and carry the 4.

carry 4
divide 60 by 8
add another 0

Dividing 40 by 8 gives 5 exactly, and the division ends (terminates). The answer to $6 \div 8$ is 0.75.

answer is 0.75
divide 40 by 8

Dividing decimals

Above, short division was used to find the decimal answer for the sum $8 \div 6$. Long division can be used to achieve the same result.

8 fits into 6 0 times, so write 0 here
multiply 8 times 0 to get 0

add decimal point
bring down a 0
divide 60 by 8

multiply 8 times 7 to get 56
first remainder is 4

bring down a 0
divide 40 by 8

First, divide 8 into 6. It goes 0 times, so put a 0 above the 6. Multiply 8×0 , and write the result (0) under the 6.

Subtract 0 from 6 to get 6, and bring down a 0. Divide 8 into 60 and put the answer, 7, after a decimal point.

Work out the first remainder by multiplying 8 by 7 and subtracting this from 60. The answer is 4.

Bring down a zero to join the 4 and divide the number by 8. It goes exactly 5 times, so put a 5 above the line.

LOOKING CLOSER

Decimals that do not end

Sometimes the answer to a division can be a decimal number that repeats without ending. This is called a "repeating" decimal. For example, here 1 is divided by 3. Both the calculations and the answers in the division become identical after the second stage, and the answer repeats endlessly.

add a decimal point to the answer line
3 does not divide into 1
carry 1
divide 10 by 3

3 goes into 10 three times, with 1 left over
divide 10 by 3

3 goes into 10 three times, with 1 left over
symbol for a repeating decimal

3 does not divide into 1, so enter 0 on the answer line. Add a decimal point after 0, and carry 1.

10 divided by 3 gives 3, with a remainder of 1. Write the 3 on the answer line and carry the 1 to the next 0.

Dividing 10 by 3 again gives exactly the same answer as the last step. This is repeated infinitely. This type of repeating decimal is written with a line over the repeating digit.